Alexandria N. Hughes, PhD

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Summary

Mathematical Statistician at the National Science Foundation, National Center for Science & Engineering Statistics (NSF/NCSES). Trained as both a statistician & neuroscientist with expertise in biological and health data science.

Education

2015 - 2020 | **PhD, Neuroscience** *University of Colorado, Anschutz Medical Campus*

2021 - 2023 | **MS, Statistics** *Texas A&M University*

2010 - 2014 | **BS, Biology; Mathematics minor** *Boise State University*

Work Experience

2022 - | **Mathematical Statistician** *National Science Foundation (NSF/NCSES)*
- Statistical consultation and review of center and external collaborative publications and projects
- Focus areas: Survey methods, STEM education & R&D, econometric models, data visualization

2021 - 2022 | **Senior Biostatistician** *Westat, Public Health & Epidemiology*
- Designed and performed statistical analyses in R in support of contracted studies; wrote scientific manuscripts and statistical analysis plans, managed data dictionaries, cloud computing (AWS)
- Focus areas: Survival analysis & Cox proportional hazards, longitudinal cohort data, mixed models

2015 - 2021 | **Graduate & Postdoctoral Training** *University of Colorado; National Institutes of Health (NIH/NICHD)*
- Acquired independent funding, published 3 first-author papers & gave 8 invited talks on projects on which I performed all experimental design, experimentation, & statistical analysis
- Computational: Daily use of R, coursework in R, Python, and MATLAB, RNA- and scRNA-seq data clustering & batch correction methods, image analysis, neuronal circuit modeling & simulation
- Biology: Genetics, molecular cloning, CRISPR, microscopy, zebrafish

2011 - 2015 | **Adjunct Faculty, Teaching Assistant, & Tutor** *Boise State University, Dept of Mathematics*
- Taught 2 sections of Elementary Algebra, TA'ed 2 sections of Calculus II and 3 sections of Calculus III, and provided drop-in tutoring services for the dept

Computational Skills

Programming	R > SAS > Python > Julia
Statistics	Regression modeling (GLM, GLMM), biostatistics (survival analysis, mixed models), Bayesian approaches (MCMC, rstan, brms), statistical/machine learning (tree methods, clustering)
Markup	LaTeX, Markdown, basic HTML/CSS
Bioinformatics	Bioconductor, RNA- and scRNA-seq analysis (scater, scran, edgeR, DEseq2) and Seurat, UCSC Genome Browser, NCBI, Ensembl, Image analysis (ImageJ)
Visualization	Data visualization (ggplot2, matplotlib, Shiny), vector graphics & scientific illustration (Inkscape)
Other	RStudio, Git, Jupyter, Quarto, writing, teaching & presenting, communicating mathematical & statistical concepts to non-technical audiences, project management and strategy

Membership & Certification

- COR (Contracting Officer's Representative) I certification
- SAS 9.4 Certified Associate
- Member, American Statistical Association

Achievements

- University of Colorado School of Medicine Dean's Outstanding Dissertation Award, 2021: Dean's selected dissertation of the 2020-2021 doctoral graduating classes
- Invited speaker at 3 high-profile conferences in neuroscience: Cold Spring Harbor Laboratory annual meetings on Glia in Health & Disease (2018) and Zebrafish Circuits & Behavior (2019) and the Gordon Research Seminar on Glial Biology (2019), as well as additional regional conferences in Colorado
- Invited seminar series speaker at New York University (2019) and Regis University (2019)
- National Science Foundation Graduate Research Fellowship Program (NSF-GRFP) awardee, 2016: funded my tuition and stipend for 3 years of my PhD

Publications

- Andreev A, Komatsu V, Almiron P, Rose K, **Hughes AN**, Lee MY (2022). Research Culture: Welcome to the lab. *eLife*, 11:e79627.
- **Hughes AN** (2022). In search of a backup career plan, I found a perfect fit. *Science* (New York, NY), 376(6589), 210-210.
- **Hughes AN** (2021). Glial cells promote myelin formation and elimination. *Frontiers in Cell and Developmental Biology*, 9, 661486.
- **Hughes AN** & Appel B (2020). Microglia phagocytose myelin sheaths to modify developmental myelination. *Nature Neuroscience*, 23, 1055–1066.
- **Hughes AN** & Appel B (2019). Oligodendrocytes express synaptic proteins that modulate myelin sheath formation. *Nature Communications*, 10(1), 1-15.
- **Hughes AN**, Oxford A, Tawara K, Jorcyk C, Oxford JT (2017). Endoplasmic reticulum stress and unfolded protein response in cartilage pathophysiology; contributing factors to apoptosis and osteoarthritis. *International Journal of Molecular Sciences*, 18(3), 665.
- **Hughes AN**, Hixon AM, Josey M (2016). A Schwanncentric View of Axon Arborization in Neuromuscular Junction (NMJ) Formation. *Journal of Neuroscience*, 36(38), 9760-9762.
- **Hughes AN** & Oxford JT (2014). A lipid-rich gestational diet predisposes offspring to nonalcoholic fatty liver disease: a potential sequence of events. *Hepatic Medicine*, 6, 15-23.